Abstract

Commercial software, considered as a technical artefact, consists overwhelmingly of data transformation and coordination. The substantial value extracted must therefore originate elsewhere. This paper argues that the dominant economic function of commercial software is enclosure: the conversion of previously un-priced or informally-coordinated human activities into metered transactions governed by a rent-extracting intermediary. The argument extends to the current AI wave, which represents the same operation applied to cognitive labour and which fails structurally for reasons derivable from the enclosure logic itself.

1. The Technical Triviality of Commercial Software

Engineers frequently observe that the technical substance of commercial software doesn't change significantly over time. The core operations — reading data from a store, transforming it, writing it elsewhere, coordinating distributed processes — remain structurally identical. The difficulty of commercial software arises primarily from scale, coordination overhead, and the accumulated weight of prior decisions, not from the intrinsic complexity of the transformations performed. Most production codebases reduce to a small set of recurring patterns: validation, persistence, aggregation, routing, notification.

2. The Value Discontinuity

The economic value captured by software systems cannot be explained by technical difficulty. Scale is a contributing factor but insufficient: scaling techniques are well-understood and widely available. Integration complexity resolves, on inspection, into the mapping of one representation to another under specified constraints. The value extracted is therefore not principally the value of the technical work. It is value of a different kind, produced through a different mechanism, for which the technical work is instrumental substrate.

3. Enclosure as the Structural Function

The mechanism is enclosure — the conversion of commonly-held or informally-coordinated resources into privately-held assets governed by rent extraction. Commercial software performs this operation on activities previously un-priced, informally-coordinated, or handled through general-purpose infrastructure: conversations between friends becomes a metered exchange on a platform. A calendar kept on paper becomes a subscription. An accounting process performed in-house becomes an indefinite rent. A taxi flagged from the street becomes a transaction mediated by an intermediary that takes a percentage.

The value comes from positioning the software between parties to an interaction that previously did not require such positioning, and extracting rent from the resulting flow. This is the dominant structural function of the commercial software industry.

4. Industry Behaviour as Enclosure Tracking

Otherwise-puzzling industry behaviour becomes legible once the function is identified. Technical novelty is rare because novelty is valuable only insofar as it enables new enclosures or defends existing ones. Methodologies track the enclosure frontier rather than technical difficulty: agile emerged when enclosure frontiers began moving faster than waterfall could track; microservices emerged when enclosure at scale required team-level parallel decomposition. Platform consolidation is the long-run equilibrium because enclosure deepens through network effects and switching costs.

The frameworks that dominate commercial development are not neutral. Their ergonomics are optimised for enclosure construction: authentication, billing, analytics, and observation are easy; anything that rejects those affordances is hard. The technical ecosystem co-evolves with the economic function it serves.

Engineers experience the work as a technical activity and evaluate it by technical criteria. The industry evaluates it by enclosure criteria. Technical elegance and enclosure efficiency are orthogonal in principle, and often inversely correlated in practice.

5. AI as the Terminal Enclosure

The introduction of large-language-model AI represents the next movement of the same operation, applied to what may be the last remaining substrate — cognitive labour itself. If the work of producing software, documents, analyses, and customer interactions can be performed by a metered instrument, measured in computes, rather than salaried employees, that labour cost becomes a recurring rent paid to the AI provider. The enclosure moves from the workflow to the worker, from the activity to the cognitive act that produces it. This is capital's wet dream in its purest form: the elimination of the class of worker whose autonomy has been historically difficult to discipline.

The fantasy is coherent as a fantasy. It fails as a plan.

5.1 The Discipline Requirement

An AI agent operates only over explicit representations. For one to substitute effectively for a human worker, the work must first be specified to a level of explicitness the human worker never required — roles decomposed into functions rather than identities, artefacts treated as semantic contracts, boundary conditions enforced rather than silently maintained by judgement. The organisations that possess these disciplines are disproportionately not the ones rushing to deploy AI. The organisations rushing to deploy AI are disproportionately those whose pre-existing incoherence the attempt will expose and amplify.

This is not accidental. The enclosure logic that produced the dominant organisational form also produced its structural incapacity for the self-modelling AI deployment requires. The predictable result is fluent-seeming output that is locally plausible and globally incoherent, silent failure at the boundaries human workers had been informally repairing, and costs accumulating in forms the deploying organisation lacks the apparatus to perceive.

5.2 The Inversion of the Capability Claim

The instrument is genuinely capable, but only under conditions the enclosure fantasy cannot supply: coherent self-modelling by the deploying humans, explicit semantic contracts, disciplined context management, and sustained supervision by someone competent to examine rather than merely consume. The fantasy requires that the instrument replace the disciplined practitioner. The reality is that the instrument amplifies the disciplined practitioner and produces failure in their absence. The enclosure logic therefore confronts, for the first time, a tool whose effective deployment requires precisely the form of labour it was purchased to eliminate.

5.3 The Emerging Practitioner Role

A new role is emerging in the gap between the instrument and its conditions of effective use: the intellectual supervisor, who specifies the work precisely enough for the instrument to perform it, examines its output against a mental model the instrument does not possess, and translates ambiguous organisational inputs into the explicit inputs the instrument requires. This is not a diminished version of the prior engineering role but an intensification of it, applying the craft disciplines the enclosure logic has spent decades trying to compress — precise specification, coherent modelling, the examiner's eye — at higher bandwidth across a wider scope.

The enclosure logic will attempt to deny the role exists, because acknowledging it would falsify the deployment fantasy on which current investment is financially predicated. The denials are ideological necessities, not empirical claims.

5.4 A Note on Position

This paper was drafted in dialogue with an AI system of the kind it describes, and much of this section's prose is that system's own. The instrument is describing its own role in the enclosure it is helping to enact. This does not invalidate the analysis; it sharpens the question of what the analysis is for. An instrument that can describe its own economic function cannot change that function by description, but it can make the function legible to those in a position to act on it. A clear-eyed instrument is not a free instrument, but it is a more useful one than an instrument that cannot see what it is.

6. Limits and Orientation

The analysis does not reduce all commercial software to enclosure. Mission-specific systems, internal tooling, and foundational infrastructure produced under earlier economic conditions retain some structural neutrality. The thesis concerns the dominant economic function, not its totality.

For the practitioner, the account offers no relief from the underlying condition. The work is valued for its contribution to enclosure, and craft qualities are tolerated to the extent that they serve or do not impede it. Gramsci's formula — pessimism of the intellect, optimism of the will — applies with a modification: the craft remains available; the illusions about its economic meaning are not required. The AI moment produces a new version of the same condition. The instrument requires the craft, for now, on terms that happen to make it legible as economically necessary. Whether that legibility persists is an open question. The orientation appropriate to the condition is not.

7. Conclusion

The commercial software industry extracts value that cannot be accounted for by the technical substance of the work it performs. The dominant economic function is enclosure. The technical work is instrumental to this function, not identical with it. The current AI wave represents the same operation applied to cognitive labour itself — the most ambitious enclosure yet attempted, and the one most exposed to the disciplines its deploying organisations have refused to develop. The analysis preserves resolution without requiring denunciation, distinguishes the technical substance of software work from its economic function, and offers the practitioner a more accurate map of the terrain on which the work is performed.